

Solutions In-Class Group Exercise

Labour Markets and the Future of Work Exercise

ECON 204 – Contemporary Economic Issues

Part I: Structural Change in Employment

Question 1

Manufacturing experienced the largest decline in employment share.

Question 2

Manufacturing fell from 25% to 10%.

$$25\% - 10\% = 15 \text{ percentage points}$$

Answer: 15 percentage points.

Question 3

The table illustrates **structural transformation**. Over time, advanced economies tend to shift:

- from agriculture,
- to manufacturing,
- and eventually toward services.

This pattern occurs because productivity growth reduces labour demand in goods-producing sectors while rising incomes increase demand for services.

Question 4

Two key explanations for the decline in manufacturing employment are:

1. **Automation and technological change.** Modern machinery and robotics allow firms to produce more output with fewer workers.
2. **Globalization and offshoring.** Labour-intensive production stages have moved to countries with lower labour costs.

Additional factors may include rising productivity and changing consumer demand toward services.

Question 5

Service-sector employment has become dominant because:

- rising incomes increase demand for services,
- productivity growth reduces labour demand in agriculture and manufacturing,
- many service occupations involve interpersonal or non-routine tasks,
- aging populations increase demand for health and care services.

Part II: Automation, Tasks, and Occupations

Question 6

The occupations most vulnerable to automation are those dominated by routine tasks.

- Data Entry Clerk — highly routine and easily automated
- Taxi Driver — vulnerable due to autonomous vehicle technologies

Moderately vulnerable occupations include accountants because some tasks can be automated.

Less vulnerable occupations include nurses and software developers, which require interpersonal interaction and complex problem solving.

Question 7

Routine tasks are easier to automate because they follow predictable and rule-based procedures. Computers and algorithms perform well when tasks involve repetitive steps that can be coded into software.

Examples include:

- bookkeeping
- data processing
- assembly-line tasks

Question 8

Tasks requiring interpersonal skills, creativity, and judgment are harder to automate because they involve:

- empathy and human interaction
- adapting to unexpected situations
- complex decision making
- creativity and innovation

Examples include teaching, nursing, and management.

Question 9

Automation often replaces specific tasks rather than entire occupations. Most jobs involve multiple tasks, some routine and some non-routine. Technology may automate the routine parts while workers continue performing tasks requiring judgment, communication, and creativity.

Part III: Wage Inequality and Skill Demand

Question 10

Wage gap in 2000:

High-skill wage: \$30/hour

Low-skill wage: \$18/hour

$$30 - 18 = 12$$

Answer: \$12 per hour.

Question 11

Wage gap in 2025:

High-skill wage: \$52/hour

Low-skill wage: \$22/hour

$$52 - 22 = 30$$

Answer: \$30 per hour.

Question 12

The wage gap increased from \$12 to \$30 per hour. This indicates that wage inequality between high-skill and low-skill workers has increased.

Question 13

High-skill workers

Initial wage: \$30

Final wage: \$52

$$52 - 30 = 22$$

$$\frac{22}{30} \times 100 = 73.33\%$$

Increase: approximately 73.3%.

Low-skill workers

Initial wage: \$18

Final wage: \$22

$$22 - 18 = 4$$

$$\frac{4}{18} \times 100 = 22.22\%$$

Increase: approximately 22.2%.

Question 14

Two explanations for the widening wage gap include:

- **Skill-biased technological change:** technology complements skilled workers and raises their productivity.
- **Rising demand for education and specialized skills:** modern economies reward workers with advanced technical and analytical abilities.

Other explanations may include globalization, declining unionization, and labour market polarization.

Part IV: Canadian Labour Market Data

Question 15

Youth workers (ages 15–24) had the highest unemployment rate in both years.

Question 16

Youth unemployment increased from 11.0% to 13.5%.

$$13.5 - 11.0 = 2.5$$

Answer: 2.5 percentage points.

Question 17

Young workers are more vulnerable to labour market disruptions because:

- they have less work experience,
- they often hold temporary or entry-level jobs,
- they are concentrated in sectors sensitive to economic cycles,
- they have less bargaining power.

Question 18

Technological change affects age groups differently.

Younger workers may adapt more easily because they possess stronger digital skills and greater flexibility.

Older workers may face higher retraining costs and may find occupational transitions more difficult late in their careers.

Part V: The Gig Economy

Question 19

Advantages of gig work include:

- flexible working hours,
- easy entry into the labour market,
- opportunities for supplemental income.

Question 20

Disadvantages include:

- lack of job security,
- absence of employer-provided benefits,
- income volatility.

Question 21

A strong answer recognizes trade-offs.

Treating gig workers as employees would provide labour protections such as minimum wages and benefits. However, it could reduce flexibility and increase costs for firms.

A hybrid approach may combine flexibility with basic protections.

Question 22

Possible policy responses include:

- portable benefits systems,
- minimum earnings standards,
- improved worker classification rules,
- collective bargaining rights,
- greater transparency in platform algorithms.

Part VI: Artificial Intelligence and Work

Question 23

AI complements labour when it increases worker productivity instead of replacing workers. Technology assists workers in completing tasks more efficiently.

Examples include AI diagnostic tools assisting doctors or AI software helping programmers write code.

Question 24

Example: doctors using AI to analyze medical scans and improve diagnostic accuracy.

Question 25

Example: data entry clerks, where repetitive tasks can be automated using software and artificial intelligence.

Question 26

Skills likely to become more valuable include:

- analytical thinking,
- problem solving,
- communication skills,
- creativity,
- adaptability,
- digital literacy.

These skills complement new technologies and are harder to automate.

Part VII: Policy Debate

Question 27

Policy effects depend on the chosen option.

For example, retraining programs can help displaced workers acquire new skills and transition to growing sectors.

Question 28

Firms may benefit from a more skilled labour force, but they may also face higher costs depending on the policy.

Question 29

Possible unintended consequences include:

- high fiscal costs,
- mismatch between training and labour demand,
- reduced hiring if labour regulations become too restrictive.

Question 30

A strong recommendation clearly states a position and explains the economic reasoning behind it while acknowledging trade-offs.

Key Takeaways

Students should understand that:

- labour markets evolve through structural transformation,
- automation often replaces tasks rather than entire jobs,
- technological change can increase wage inequality,
- gig work introduces flexibility but also insecurity,
- artificial intelligence may both complement and substitute labour,
- public policy plays an important role in shaping future labour market outcomes.